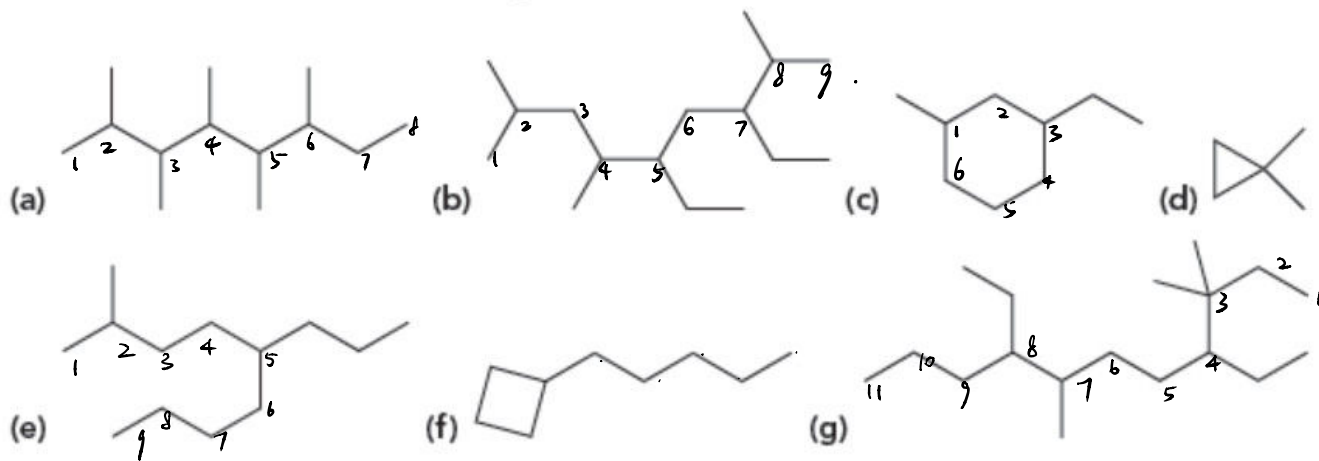


第二章作业（手写！）

陈天乐
无25
2022/10/8

1、用系统命名法命名下列分子：



(a). 2,3,4,5,6-五甲基辛烷

(c). 2-甲基-5-丙基环己烷

(b). 2,3,8-三甲基-5,7-二乙基壬烷

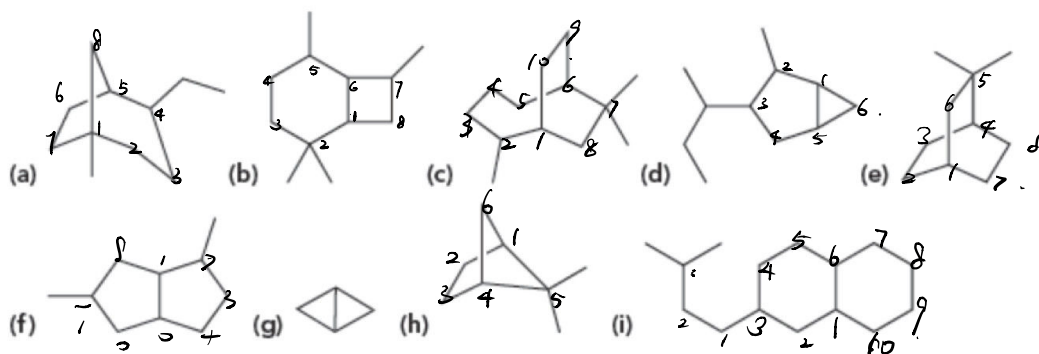
(f). 环丁基戊烷

(e). 1-甲基-3-乙基环己烷

(g). 3,3,7-三甲基-4,8-二乙基十一烷

(d). 1,1-二甲基环丙烷

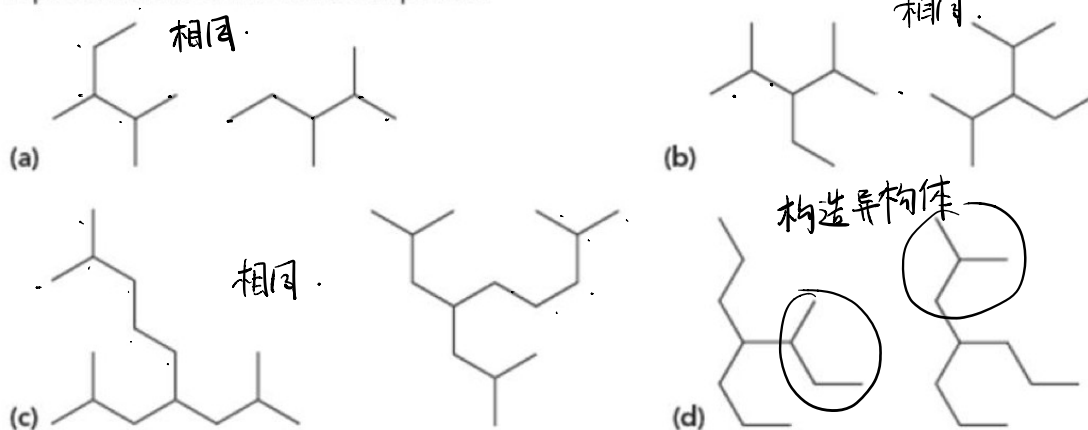
2、用系统命名法命名下列分子：



- (a). 1-甲基-4-乙基二环[3.2.1]辛烷. (b). 2,2,5,7-四甲基二环[4.2.0]辛烷.
 (c). 2,7,7-三甲基二环[4.2.2]癸烷. (d). 2-甲基-3-异丁基二环[3.1.0]己烷.
 (e). 5,5-二甲基二环[2.2.2]辛烷. (f). 2,7-二甲基二环[3.3.0]辛烷.
 (g). 二环[1.1.0]丁烷. (h). 5,5-二甲基二环[2.1.1]己烷.
 (i). 3-3-二甲基二环[4.4.0]癸烷.

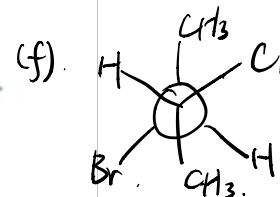
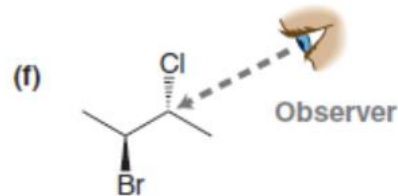
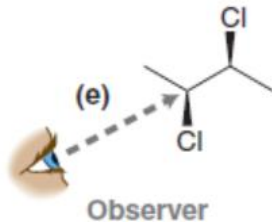
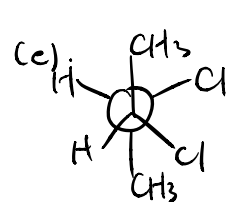
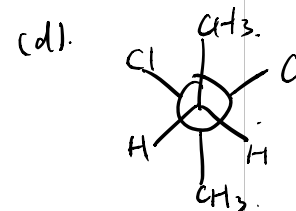
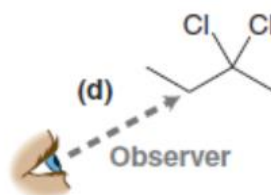
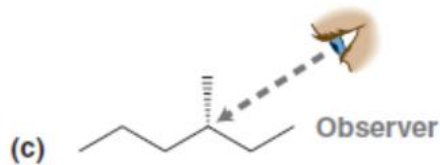
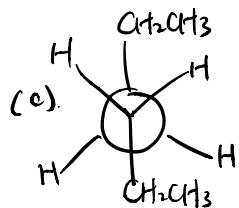
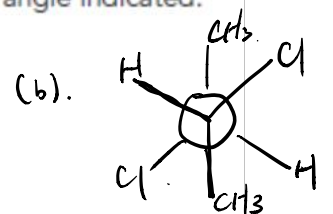
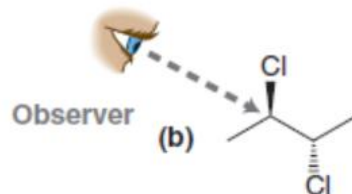
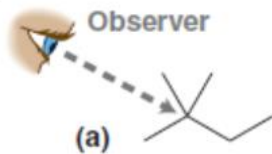
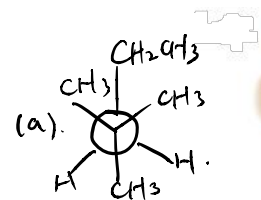
3、构造异构体还是相同的分子？

For each pair of compounds, identify whether they are constitutional isomers or two representations of the same compound:



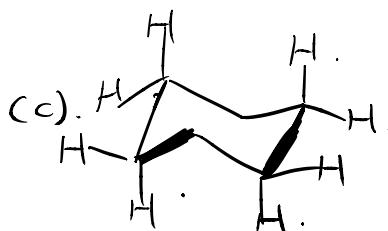
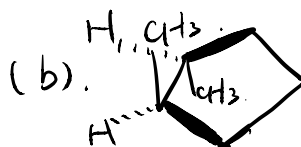
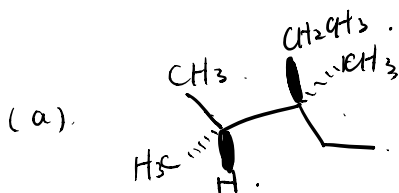
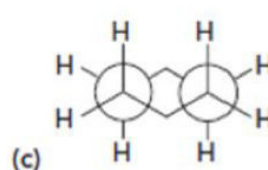
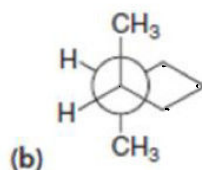
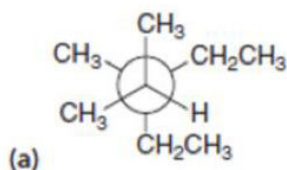
4、画出下列分子的Newman投影式

In each case below, draw a Newman projection as viewed from the angle indicated:



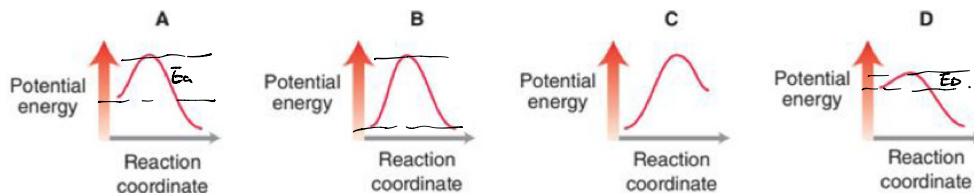
5、画出下列Newman投影式的键线式表示形式

Draw a bond-line structure for each of the following compounds:



6、比较反应的能量进程图，回答问题

Consider the relative energy diagrams for four different processes:



- Compare energy diagrams A and D. Assuming all other factors (such as concentrations and temperature) are identical for the two processes, identify which process will occur more rapidly. Explain.
- Compare energy diagrams A and B. Which process will more greatly favor products at equilibrium? Explain.
- Do any of the processes exhibit an intermediate? Do any of the processes exhibit a transition state? Explain.
- Compare energy diagrams A and C. In which case will the transition state resemble reactants more than products? Explain.

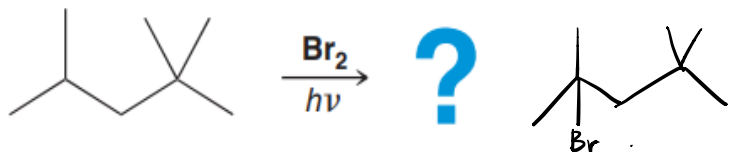
- Compare energy diagrams A and B. Assuming all other factors (such as concentrations and temperature) are identical for the two processes, identify which process will occur more rapidly. Explain.
- Compare energy diagrams B and D. Which process will more greatly favor products at equilibrium? Explain.
- Compare energy diagrams C and D. In which case will the transition state resemble products more than reactants? Explain.

- (a). 由图可知 $E_A > E_D \rightarrow$ D 过程发生更快
- (b). 在平衡时, 由于 A 过程放出热量更多, 并且活化能更低, 更有利于反应正向进行。
- (c). 4 个阶段都没有中间体, 但都有过渡态。因为能量变化中没有低谷, 只有山峰。

- (d). 根据 Hammond 假设, A 反应放热 $\rightarrow$ 平衡点更靠近反应物, C 反应吸热 $\rightarrow$ 平衡点更靠近产物, 故 B 更像反应物。
- (e). 由图可知 $E_A < E_D \rightarrow$ A 过程发生更快
- (f). 在平衡时, 由于 D 反应放热更多, 且 $E_D < E_B$, 更有利于反应正向进行。
- (g). 根据 Hammond 假设, D 反应放热 $\rightarrow$ 平衡点更靠近反应物, C 反应吸热 $\rightarrow$ 平衡点更靠近产物, 故 B 更像产物。

7、

Predict the major product obtained upon radical bromination of 2,2,4-trimethylpentane:



8、 Compound A has molecular formula C_5H_{12} and undergoes ^{氯代}monochlorination to produce four different constitutional isomers.

(a) Draw the structure of compound A. (a). 只有 2-methylbutane 有四种一氯代物.

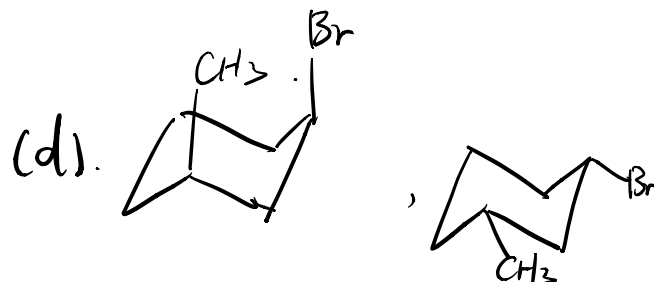
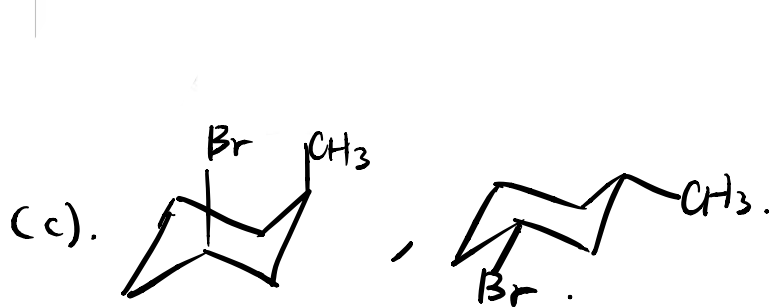
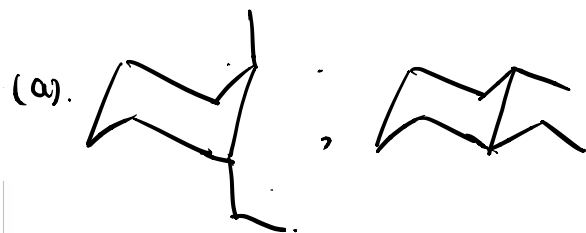
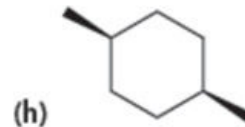
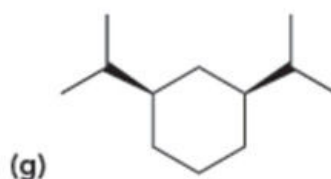
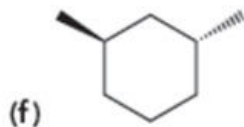
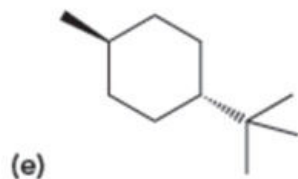
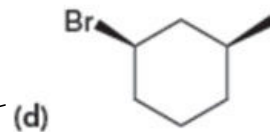
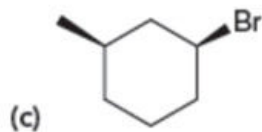
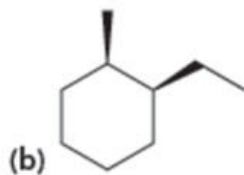
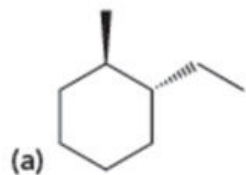
(b) Draw all four monochlorination products. (b).

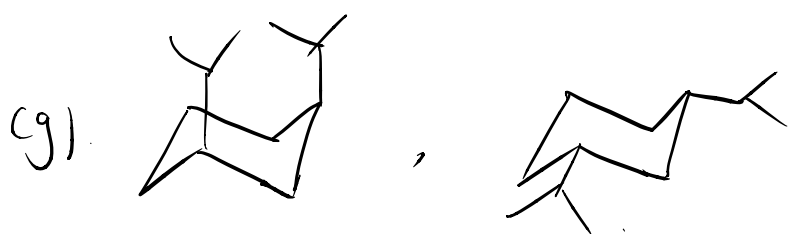
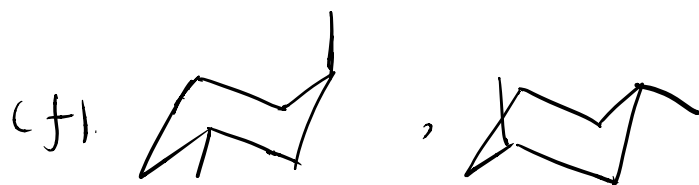
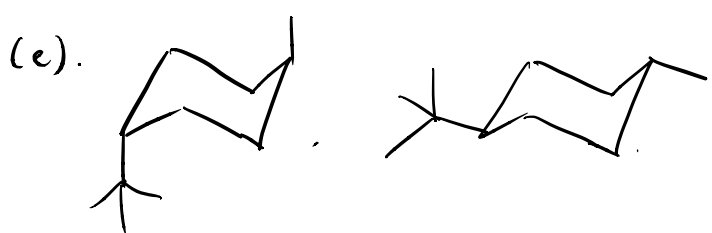
(c) If compound A undergoes ^{溴代}monobromination (instead of monochlorination), one product predominates. Draw that product.

(c). ^{最稳定}_{中间件} : ← Dominant Product .

9.

Draw both conformations for each of the following compounds:





10. Draw the lowest energy conformation for each of the following compounds:

